

6CCSAHAI

Human AI Interaction

LECTURE 2 – LGT

Cognitive Psychology and Mental Models in AI Design

[What do we want to get done today ?]

1. Kahoot-like Quiz and questions [15 min]

2. LGT-task: [30 min]

- analyse a gen-AI system for distributed cognition & CASA elements
- create coherent conceptual / mental models.

3. SGT-task transition: [45 min + SGT]

- develop your own system + test with colleagues

Teaching Assistant (TA)



Dariana Dorin

Bsc in Computer Science — King's College London
Internships at BMW, Amazon, UCL

- The TA is an expert who has experience in HAI
- She will be running the Small Group Tutorials (SGTs)

Go to
www.menti.com

Enter the code

6699 6383

Call me on whatsapp on me you in the car?

Join at menti.com | use code **6699 6383**



Or use QR code



PS

Menti

HAI-LGT-Lect2



Choose a slide to present



Quiz leaderboard

No results yet

Top Quiz participants will be displayed here once there are results!

Conceptual model is something actively created by the user to understand how your system works

[theme of the day]

Computational offloading ... aka making cognitive tasks easier

*Machines store our thoughts,
Our minds freed to wonder more—
Technology aids.*

*There once was a brain feeling fried,
With tasks piling up, it just sighed,
"Let computers assist,
Cross things off my list!"
Now thinking feels less like a ride.*

[links gdoc]



*Machines store our thoughts,
Our minds freed to wonder more—
Technology aids.*

*There once was a brain feeling fried,
With tasks piling up, it just sighed,
"Let computers assist,
Cross things off my list!"
Now thinking feels less like a ride.*

[LGT task – work in pairs]

Your mission: micro-narrative design analysis

Step 1:

Take up to 15 minutes to go through the micro-narrative experience and note down instances of:

- all design elements that rely on distributed cognition aspects
- all design elements that could be investigated through CASA

At the same time, keep track of the **mental model** that you as participants develop about the system.

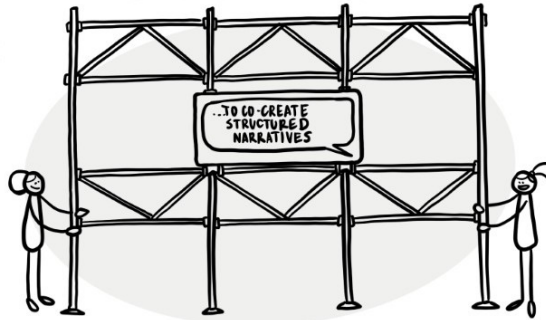
Draw it on a piece of paper – we will discuss together

CO-CREATES
STORIES

TURNING SHARDS OF THOUGHT...



PROVIDES
SCAFFOLDING...



HELPS
ANSWER
RESEARCH
QUESTIONS



PROVIDES A MORE
ENGAGING WAY OF
GATHERING
DATA



MICRO NARRATIVES WHAT IT CAN DO...

SHOWS
PATTERNS

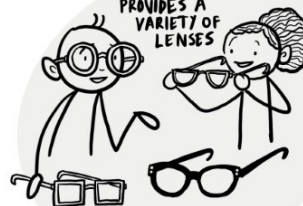


ALLOWS
SPEED and SCALE

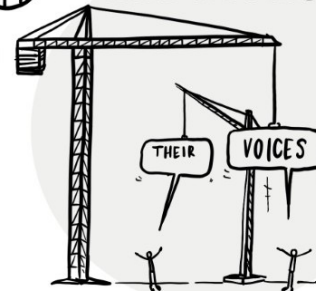


PROVIDES A
VARIETY OF
LENSES

HELPS PEOPLE
SEE THINGS
DIFFERENTLY



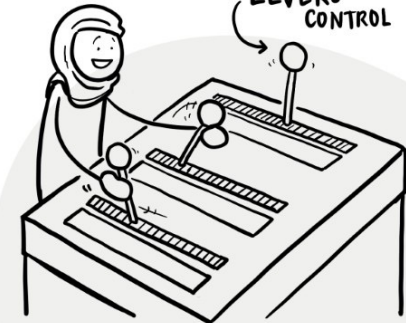
ENGAGES PEOPLE
AND ELEVATES



CHALLENGES THE
POWER
DYNAMICS



... GIVING PARTICIPANTS
THE LEVERS OF
CONTROL



[LGT task – work in pairs]

Your mission: micro-narrative design analysis

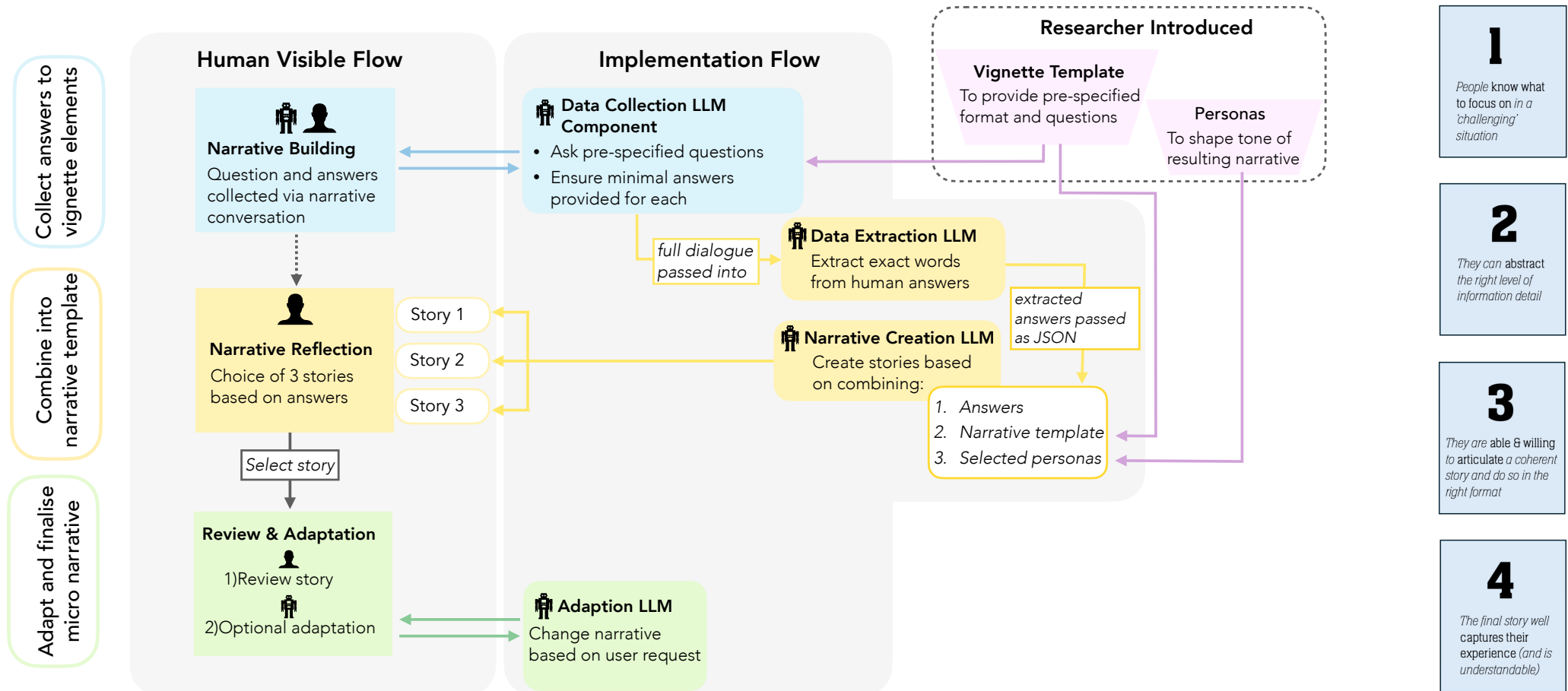
Step 2:

Now compare your mental model with the conceptual model of the system as described on this diagram on the next slide.

- What are the main differences?
- Are there any distributed cognition design elements that you missed?
- What does this mean for you as designers if you will be building your own micro-narrative

Take up to 15 minutes for this task.

Design overview – “orchestrated cognition delegation & scaffolding”



[SGT task – work in pairs]

Mission: design a new micro-narrative to collect these stories

*Tell me about a time when you really understood something
difficult as part of an LGT.*

Tell me about a time when you really understood something difficult as part of an LGT.

Role: Educational specialist, collecting 100s data point from university students

Output: Two files to submit

1. Ready-to-test config file to test in our demo environment

2. One-page explanation of:

- Questions / template choices
- Personas (students' voice, teacher-facing, 'cheeky/fun')

[SGT task – work in pairs]

Step 1 – in LGT:

Think about the types of questions / knowledge you'd want to include in the story.

You can

- work with the 'bot factory' to test your possible questions
- Mistral / LLMs to help you understand the design space

[SGT task – work in pairs]

Step 2 – in SGT:

Finalise your system & config file & one-pager.

You are encouraged to prototype quickly, and ideally also work with other teams to stress test each others' prototypes later in the process

Heads up for next week – I'll be teaching remotely from MIT



People

Pat Pataranutaporn

Assistant Professor of Media Arts and
Sciences

Pattie Maes

Professor of Media Technology;
Germeshausen Professor

Valdemar Danry

Graduate Student

Auren Liu

Graduate Student (HST)

Projects

Moonshot: Benchmarks for Human
Flourishing with AI

Tuesday — Wednesday

October 14, 2025 —

October 15, 2025

MIT Media Lab,
Cambridge, MA

Collaborative Workshop on Benchmarks for Human Flourishing with AI

Workshop dates: October 14-15, 2025

Deadline for application : August 1, 2025



Welcome



84 Participants | 47 from Academia, 22 from Industry, 13 from Non-profits, 2 Government

Join at menti.com | use code **1217 9973**



This LGT was useful for me

This LGT was well prepared

This LGT was enjoyable

Strongly disagree

Strongly agree

